

Thesis Structure (6-Chapter Model)

Randomized Reward Shaping Deep Reinforcement Learning
for Self-Adaptive Systems

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1.2.2 Motivating Case Study: The DeltaIoT System (A First Look)

1.2.3 Sources and Impact of Uncertainty in DeltaIoT

1.3 Learning-Based Adaptation: The Promise and Pitfalls of Reinforcement Learning

1.3.1 Why RL for Runtime Decision-Making?

1.3.2 Limits of Classic RL in Large, Discrete Adaptation Spaces

1.4 Problem Statement and Research Questions

1.5 Proposed Solution: Reward-Shaped Deep Reinforcement Learning (RS-DRL)

Integrated within the MAPE-K (Monitor-Analyze-Plan-Execute-Knowledge) feedback loop architecture.

1.5.1 Core Idea: Randomized Reward Reshaping (ρ -factor) during Experience Replay

Controlled by a reshaping factor ρ determining the proportion of failed experiences to reshape.

1.6 Key Contributions of the Thesis

1.7 Thesis Outline

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4.3.1 Core Algorithm: DQN for Streaming Adaptation Cycles

Addressing the limits of classic RL through continuous, streaming learning interaction.

4.3.2 The Reward Shaping Function (Algorithm 2)

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